

DIP BHATTACHARJEE

Computer Science & Engineering Undergraduate | AI/ML Research Enthusiast

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate majoring in **Computational Theory**, with a strong interest in Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, and applied research. Academic training includes Machine Learning, Natural Language Processing, Graph Theory, Data Structures and Algorithms, and Object-Oriented Programming. Experienced in programming, software development, AI/ML-based projects, and solving computational problems. Passionate about developing practical, data-driven solutions and continuously expanding technical expertise in AI/ML, NLP, and emerging intelligent technologies.

EDUCATION

Bachelor of Science in Computer Science and Engineering

American International University-Bangladesh

Aug 2023 – Present

CGPA: 3.82/4.00

Major: Computational Theory

Completed Major Courses: Natural Language Processing, Machine Learning, Basic Graph Theory.

Additional Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming with Java, Programming in C++, and Discrete Mathematics.

Higher Secondary Certificate (HSC)

Mohipal Government College – Science

2020 – 2022

GPA: 4.89/5.00

Secondary School Certificate (SSC)

Batisa Secondary School – Science

2018 – 2020

GPA: 4.89/5.00

RESEARCH & CONFERENCE PAPERS

Progressive Two-Stage Transfer Learning with Dual Imbalance Compensation for Interpretable Enamel Caries Classification

Accepted for Presentation at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)

Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System

Accepted for Presentation at IEEE TENSYP 2026

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, C#, JavaScript

Artificial Intelligence: Machine Learning, Deep Learning Fundamentals, Transfer Learning, Classification, Model Training and Evaluation

Natural Language Processing: NLP Fundamentals, Text Preprocessing, Text Classification, Feature Extraction and Language Processing Concepts

Data & ML Processing: Data Preprocessing, Data Analysis, Model Evaluation, Handling Class Imbalance

Core Computer Science: Data Structures and Algorithms, Basic Graph Theory, Object-Oriented Programming, Problem Solving

Research Areas: Machine Learning, Natural Language Processing, Computer Vision, Medical Image Analysis, Explainable AI, IoT-based Intelligent Systems

Applications & Tools: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, AutoCAD, Adobe Photoshop

PROJECTS

Hospital Management System

Java

- Developed a system to support essential hospital operations, including patient records, appointments, billing, and staff management.
- Applied object-oriented programming principles in Java to design structured and maintainable system functionality.

Poultry Farm Management System

Java, Database

- Developed a database-driven management system to organize and manage poultry farm-related information efficiently.
- Applied Java programming and database concepts to implement structured data storage, management, and system operations.

EXPERIENCE

Instructor

NextGen Coaching Center for Admission Preparation

- Taught key academic subjects required for university admission preparation, helping students strengthen fundamental concepts.
- Developed mock tests and quizzes to improve students' problem-solving speed, accuracy, and examination preparation.
- Conducted one-on-one mentoring sessions to help students understand challenging topics and improve their academic confidence.

Freelance Tutor

Programming & Problem Solving

- Mentored students in Python, JavaScript, and C++, strengthening programming fundamentals and problem-solving skills.
- Designed personalized learning plans covering programming, data structures, algorithms, and computational problem solving.
- Guided students through practical coding exercises and project-based learning activities.

ACHIEVEMENTS & CERTIFICATIONS

- Dean's Award

ADDITIONAL INFORMATION

Languages:

Bangla (Native), English (Fluent), Hindi (Proficient in Listening, Basic Speaking)

Research Interests:

Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Explainable AI, Medical Image Analysis, and Intelligent Systems